

ML Basics& Linear Regression

QUESTION 1

What is Machine Learning? Explain the main types of Machine Learning with examples.

ANSWER

Machine Learning is a branch of Artificial Intelligence (AI) that allows computers to learn from data and make decisions or predictions without being explicitly programmed.

Input Data + Rules (Program) → Output

Machine Learning is used to:

Predict future outcomes

Find hidden patterns in data

Automate decision-making

Improve accuracy over time

Handle large amounts of data

Main Types of Machine Learning

Supervised Learning

Unsupervised Learning

Reinforcement Learning

1) Supervised Learning

Supervised learning is a type of machine learning where the model learns from labeled data.

Labeled data means:

Input + Correct Output is already provided.

The model learns the relationship between:

House Features → Price

Then predicts the price of a new house.

2) Unsupervised Learning

Unsupervised learning works with unlabeled data.

The model finds hidden patterns or groups by itself.

		The model creates groups:								
		Group 1 → High spending customers								
		Group 2 → Low spending customers								
		3) Reinforcement Learning								
		Reinforcement Learning is a type of ML where an agent learns by interacting with an environment and receives rewards or penalties.								
		It learns through trial and error.								
		Examples:								
		Self-driving cars								
		Game playing AI								
		Robotics								
		Trading systems								

QUESTION 2	What is Supervised Learning? Explain the difference between Regression and Classification.
ANSWER	Supervised Learning is a type of Machine Learning where a model learns from labeled data (data that already has the correct output).
	In supervised learning, we provide:
	Input Data (Features) + Correct Output (Label) → Model learns patterns → Predicts new output
	The ML model learns:
	Hours + Attendance → Result
	Supervised Learning is mainly divided into:
	Regression
	Classification
	1) Regression
	Regression is a supervised learning technique used to predict continuous numerical values.
	The output is a number.
	Examples:
	House Price Prediction
	2) Classification
	Classification is a supervised learning technique used to predict categories or classes.
	The output is a label/category.
	Examples:
	Email Spam Detection
	Input:
	Email content

QUESTION 3

What is Linear Regression? Write the Linear Regression equation and explain intercept and coefficient

ANSWER

Linear Regression is a supervised machine learning algorithm used to predict a continuous numerical value by finding the relationship between input variables (features) and an output variable (target).

It assumes that there is a linear relationship between input and output.

Example:	Predicting house price:
-----------------	-------------------------

Input:	Area	
	Number of bedrooms	
	Location	

Output:	House Price
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The model finds a line that best fits the data and uses that line to make predictions.

Where:	$Y=b_0+b_1X$
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Y = Predicted Output (Dependent Variable)

The value we want to predict.

Examples:

House price
Salary
Sales

X = Input Feature (Independent Variable)

The variable used to make prediction.

Examples:	Area of house	
	Years of experience	
	Advertising cost	

b_0 = Intercept

Intercept is the value of Y when X is zero.
It is the starting point of the regression line.

Example:

Equation:					
Salary=20000+5000(Experience)Salary = 20000 + 5000(Experience)Salary=20000+5000(Experience)					

b_1 = Coefficient (Slope)

Coefficient shows how much Y changes when X increases by 1 unit.

Example:

Equation:				
Salary=20000+5000(Experience)Salary = 20000 + 5000(Experience)Salary=20000+5000(Experience)				

[illegible]

QUESTION 4		Explain the following evaluation metrics used in Linear Regression: SSE MSE RMSE R² Score Adjusted R²						
ANSWER		1) SSE (Sum of Squared Errors) SSE is the sum of the squared differences between actual values and predicted values.						SSE=Σ(Yi-Yi[^])²
		2) MSE (Mean Squared Error) MSE is the average of squared errors. It tells the average prediction error.						MSE=nΣ(Yi-Yi[^])²
		3) RMSE (Root Mean Squared Error) RMSE is the square root of MSE.						RMSE=√MSE
		4) R² Score (Coefficient of Determination) R ² measures how much variation in the target variable is explained by the model.						R²=1-SSE / SST
		Adjusted R² Adjusted R ² is a modified version of R ² that penalizes unnecessary features.						

				RMSE → because it is easy to interpret.				
				R ² Score → because it tells how well the model explains the data.				
				Adjusted R ² → preferred when comparing models with different numbers of features.				
				SSE → Total Squared Error				
				MSE → Average Squared Error				
				RMSE → Square Root of MSE				
				R ² → Percentage of variance explained by model				
				Adjusted R ² → R ² with penalty for unnecessary features				

QUESTION 5	Explain the Machine Learning workflow steps from data collection to model evaluation.		
ANSWER	A Machine Learning workflow is a step-by-step process used to build, train, test, and improve a machine learning model.		
The main steps are:			
		1) Data Collection	
Data Collection			
↓		The first step is collecting relevant data required for solving the problem.	
Data Cleaning			
↓		Data can come from:	
Exploratory Data Analysis (EDA)			
↓		Databases	
Feature Engineering		Websites	
↓		APIs	
Data Splitting		Sensors	
↓		Surveys	
Model Selection		Business records	
↓		Example:	
Model Training		For house price prediction:	
↓			
Model Testing		2) Data Cleaning (Data Preprocessing)	
↓			
Model Evaluation		Raw data usually contains errors, missing values, or duplicates. Data cleaning improves data quality.	
↓		Handling Missing Values	
Model Deployment		Removing Duplicates	
		Example:	
		Removing repeated customer records.	
		Handling Outliers	

				Example:						
				Most salaries:						
				30,000 - 80,000						
				3) Exploratory Data Analysis (EDA)						
				EDA is the process of understanding data using statistics and visualization.						
				We analyze:						
				Patterns						
				Relationships						
				Distributions						
				Correlations						
				4) Feature Engineering						
				Feature engineering means selecting, transforming, or creating useful input variables for the model.						
				Example:						
				Original data:						
				Date = 21-06-2026						
				5) Data Splitting						
				The dataset is divided into:						
				Training Data						

				Used to teach the model.					
				6) Model Selection					
				Choosing the appropriate ML algorithm based on the problem.					
				7) Model Training					
				The model learns patterns from training data.					
				8) Model Testing					
				After training, we test the model using unseen data.					
				9) Model Evaluation					
				Evaluation measures how well the model performs.					
				10) Model Deployment					
				After successful evaluation, the model is used in real applications.					

QUESTION 6

Load the Boston Housing dataset and perform basic data exploration using:

ANSWER

head()

info()

describe()

The screenshot displays a JupyterLab environment with the following components:

- Browser Tabs:** Includes 'New tab', 'Untitled33', 'Home', 'BostonHousing.csv', and 'Untitled34'.
- Address Bar:** Shows the URL 'localhost:8888/notebooks/Untitled34.ipynb?'.
- Left Sidebar (Anaconda Toolbox):** Contains sections for 'Anaconda Cloud' (with buttons for 'Create a New Project' and 'Create a New Notebook'), 'Code Snippets' (with 'Manage Code Snippets'), and 'Anaconda AI Assistant' (with 'Chat-based Python Help').
- Top Panel (JupyterLab Header):** Shows 'Untitled34' and 'Last Checkpoint: 4 minutes ago'. It includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations and code execution.
- Code Editor:** Contains the following Python code:

```
[4]: df.head()

[5]: df.describe()

[6]: df.info()
```
- Output Area:** Displays the results of the code execution:
 - df.head():** Shows the first 5 rows of the dataset with columns: crim, zn, indus, chas, nox, rm, age, dis, rad, tax, ptratio, b, lstat, medv.
 - df.describe():** Provides summary statistics for the dataset, including mean, standard deviation, and count for each column.
 - df.info():** Shows the data type and non-null count for each column, indicating 506 rows and 14 columns.

QUESTION 7

Perform Exploratory Data Analysis:

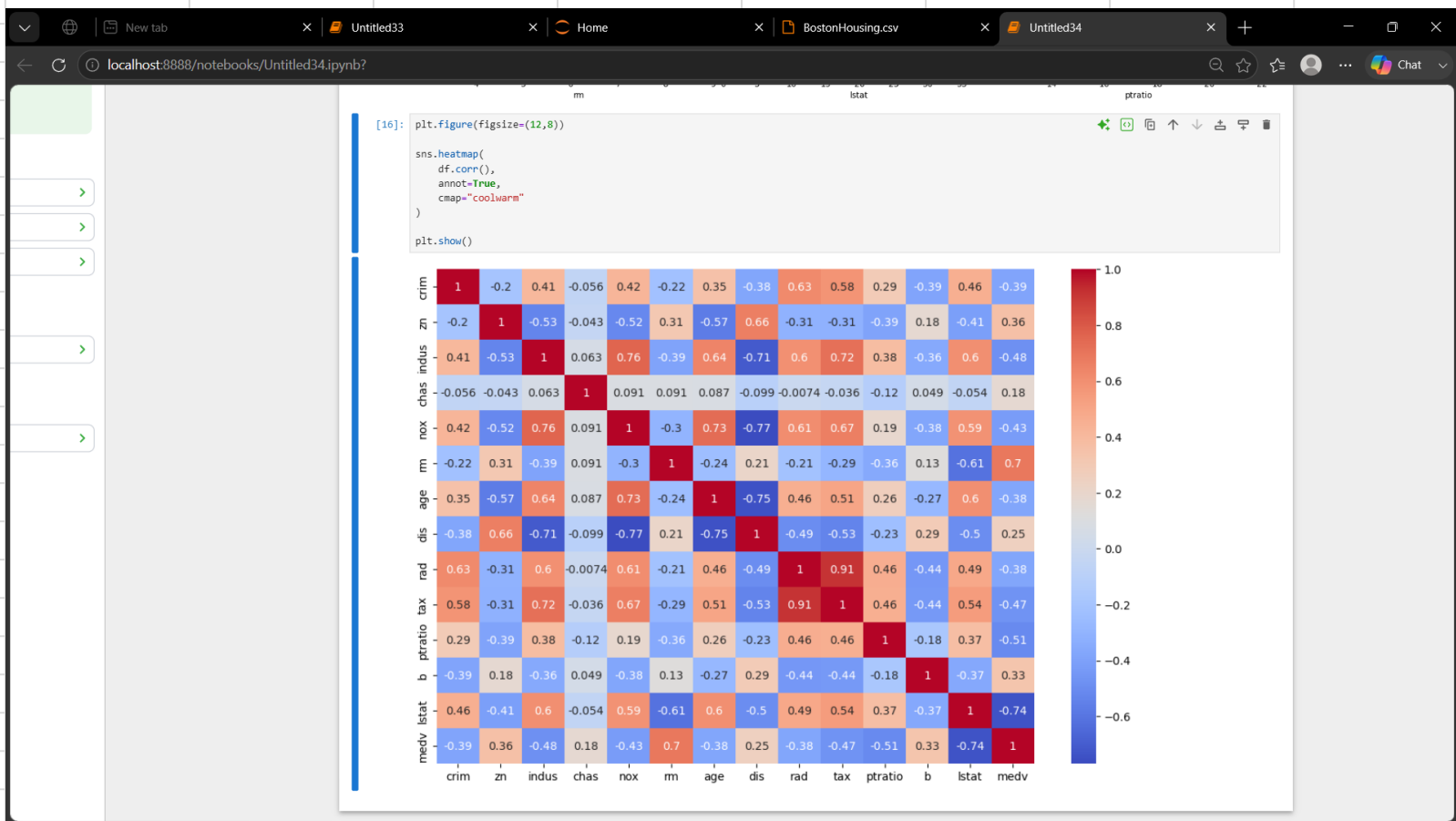
ANSWER

Create pairplot

Create correlation heatmap

Identify which features are highly correlated with house price (medv)





localhost:8888/notebooks/Untitled34.ipynb?

crimzninduschasnoxrmagedisradtaxptratio b

[18]: correlation = df.corr()['medv'].sort_values(ascending=False)

print(correlation)

medv1.000000
rm0.695360
zn0.360445
b0.333461
dis0.249929
chas0.175260
age-0.376955
rad-0.381626
crim-0.388305
nox-0.427321
tax-0.468536
indus-0.483725
ptratio-0.507787
lstat-0.737663
Name: medv, dtype: float64

Feature	Correlation	Relationship
LSTAT	-0.74	Strong negative
RM	0.69	Strong positive
PTRATIO	-0.51	Negative
TAX	-0.46	Negative
NOX	-0.43	Negative

QUESTION 8

Select independent variables (features) and dependent variable (target).

Target variable: med

ANSWER

```
[23]: X = df.drop('medv', axis=1)
      X.head()
```

```
[23]:
```

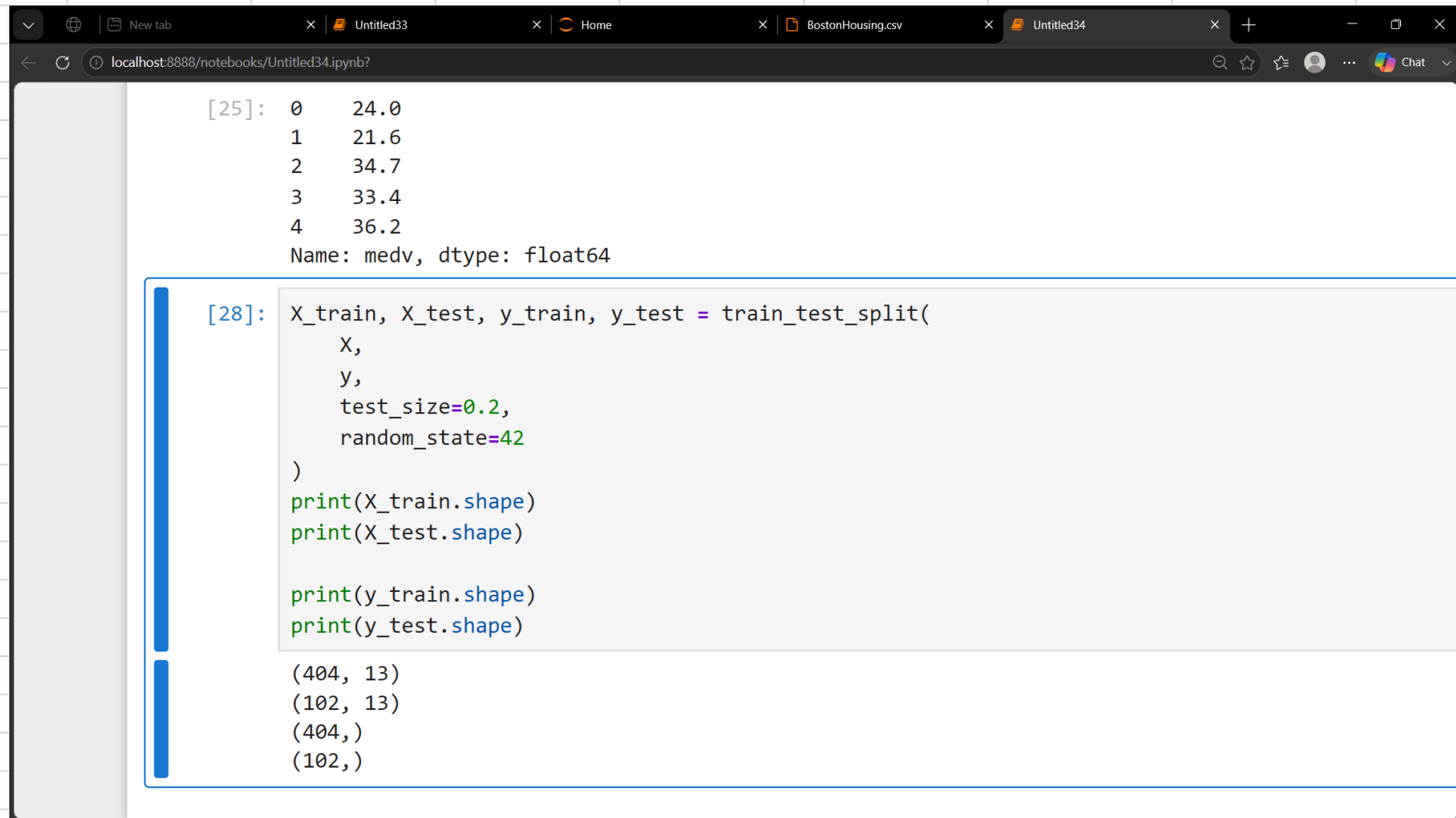
	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	lstat
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.98
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.14
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.03
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.33

```
[25]: y = df['medv']
      y.head()
```

```
[25]: 0    24.0
      1    21.6
      2    34.7
      3    33.4
      4    36.2
      Name: medv, dtype: float64
```

QUESTION 9

Split the dataset into training and testing sets (80% training, 20% testing).

ANSWER

The screenshot shows a Jupyter Notebook window with several tabs: 'New tab', 'Untitled33', 'Home', 'BostonHousing.csv', and 'Untitled34'. The active tab is 'Untitled34', displaying a Jupyter Notebook interface. The notebook has a dark theme. The first cell, labeled '[25]:', contains a list of five data points with their corresponding 'medv' values: 0 (24.0), 1 (21.6), 2 (34.7), 3 (33.4), and 4 (36.2). The data type is identified as 'float64'. The second cell, labeled '[28]:', contains Python code using 'train_test_split' to split the dataset into training and testing sets. The code specifies a test size of 0.2 and a random state of 42. Below the code, the shapes of the resulting arrays are printed: (404, 13) for X_train, (102, 13) for X_test, (404,) for y_train, and (102,) for y_test.

```
[25]: 0    24.0
      1    21.6
      2    34.7
      3    33.4
      4    36.2
      Name: medv, dtype: float64

[28]: X_train, X_test, y_train, y_test = train_test_split(
      X,
      y,
      test_size=0.2,
      random_state=42
      )
      print(X_train.shape)
      print(X_test.shape)

      print(y_train.shape)
      print(y_test.shape)

      (404, 13)
      (102, 13)
      (404,)
      (102,)
```


QUESTION 10

Train a Linear Regression model and perform the following:

Make predictions

Calculate MAE

Calculate MSE

Calculate RMSE

Calculate R² Score

Calculate Adjusted R²

Write the Linear Regression equation

Interpret the model coefficients

ANSWER

```
[47]: from sklearn.metrics import mean_absolute_error
      from sklearn.metrics import mean_squared_error
      from sklearn.metrics import r2_score
      import numpy as np
      model = LinearRegression()
      model.fit(X_train, y_train)

      y_pred = model.predict(X_test)
      #print('prediction', y_pred)
      #print('actual:', y_test.values)
      mae = mean_absolute_error(y_test, y_pred)
      mse = mean_squared_error(y_test, y_pred)
      rsme = np.sqrt(mse)
      r2 = r2_score(y_test, y_pred)

      n = X_test.shape[0]
      p = X_test.shape[1]
      adjusted_r2 = 1 - ((1-r2)*(n-1))/(n-p-1)
      print('r2', r2)
      print('mae', mae)
      print('rsme', rsme)
      print('adjusted_r2', adjusted_r2)
      print(model.coef_)
      print(model.intercept_)

r2 0.6687594935356329
mae 3.189091965887843
rsme 4.92860218266533
adjusted_r2 0.6198262368988514
[-1.13055924e-01  3.01104641e-02  4.03807204e-02  2.78443820e+00
 -1.72026334e+01  4.43883520e+00 -6.29636221e-03 -1.44786537e+00
  2.62429736e-01 -1.06467863e-02 -9.15456240e-01  1.23513347e-02
 -5.08571424e-01]
30.246750993924394
```